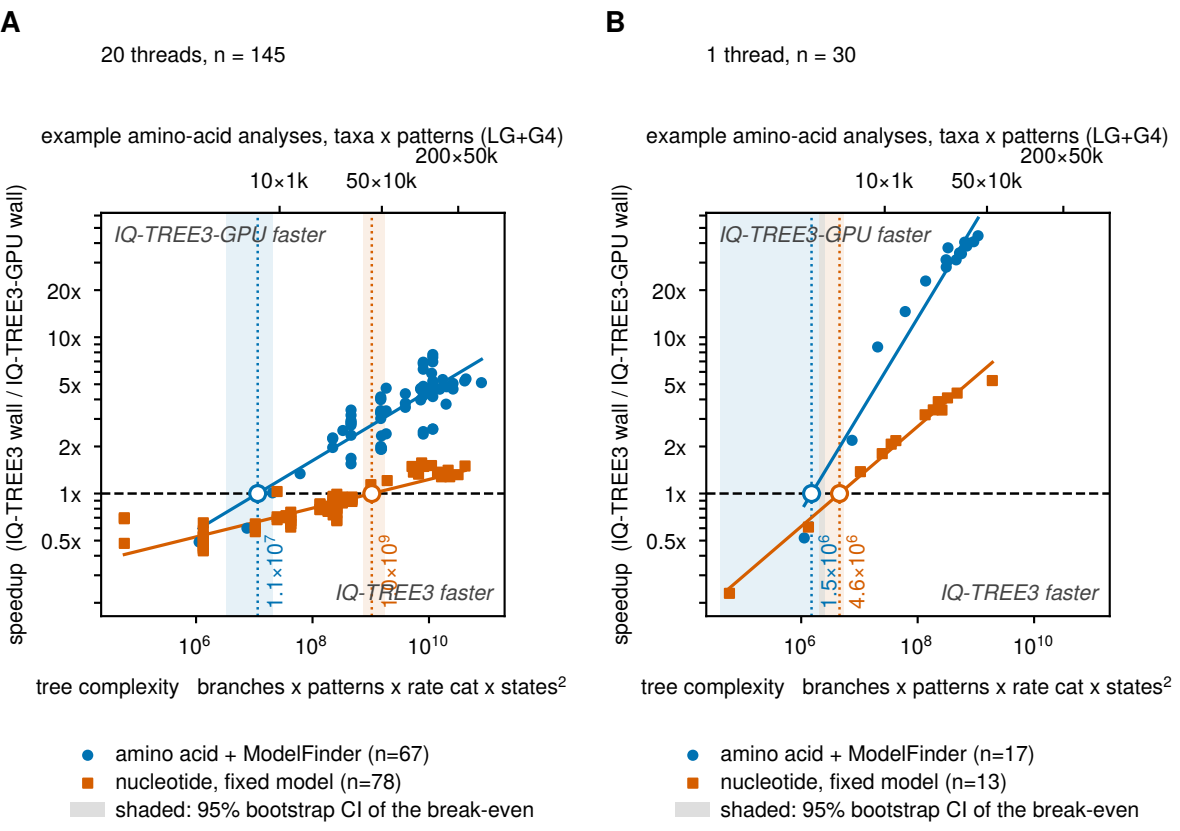

A
20 threads, n = 145
example amino-acid analyses, taxa x patterns (LG+G4)
200×50k
10×1k
50×10k
IQ-TREE3-GPU faster
IQ-TREE3 faster
20x
10x
5x
2x
1x
0.5x
speedup (IQ-TREE3 wall / IQ-TREE3-GPU wall)
1.1×10^7
1.2×10^9
10^6
10^8
10^10
tree complexity branches x patterns x rate cat x states^2
amino acid + ModelFinder (n=67)
nucleotide, fixed model (n=78)
shaded: 95% bootstrap CI of the break-even
B
1 thread, n = 30
example amino-acid analyses, taxa x patterns (LG+G4)
200×50k
10×1k
50×10k
IQ-TREE3-GPU faster
IQ-TREE3 faster
20x
10x
5x
2x
1x
0.5x
speedup (IQ-TREE3 wall / IQ-TREE3-GPU wall)
1.5×10^6
4.6×10^6
10^6
10^8
10^10
tree complexity branches x patterns x rate cat x states^2
amino acid + ModelFinder (n=17)
nucleotide, fixed model (n=13)
shaded: 95% bootstrap CI of the break-even